Software Requirements Specification

Church Collection Management System

Version 1.0 · May 2026

Classification: Confidential

Architecture: Layered Monolithic · Platform: JavaFX / MySQL

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1. Introduction

> *This document defines the complete software requirements for the Church Collection Management System — covering architecture, functional requirements, security, and deployment specifications.*

1.1 Purpose

The purpose of this system is to manage weekly church collection submissions, receipt generation, reporting, user management, and auditing for multiple churches under multiple regions. The system is designed as a standalone desktop application with future capability for multi-user operation over a Local Area Network (LAN).

1.2 Scope

The system will address the following functional areas:

- Manage churches and regions
- Record weekly collection submissions
- Generate and manage receipts
- Maintain submission statuses and tracking
- Generate comprehensive reports
- Manage users, roles, and permissions
- Maintain audit and activity logs
- Support manual and automatic database backups
- Support future centralized database deployment

1.3 Intended Users

User Role	Description
Administrator	Full system access; manages users, receipts, configuration
Data Entry Operator / User	Limited access for daily data entry operations
Regional Supervisor	Future role — regional oversight and reporting
Auditor	Future role — view-only access for audit purposes

2. System Overview

2.1 Application Type

Desktop Application — built for Windows, with a layered monolithic architecture supporting both standalone and future LAN deployments.

2.2 Architecture Style

Layered Monolithic Architecture ensuring clear separation of concerns across presentation, business logic, and data access layers.

2.3 Deployment Modes

Mode	Description
Initial Deployment	Single Windows machine, standalone operation
Future Deployment	Multiple client machines over LAN with centralized database server

3. Technology Stack

The following technologies have been selected to ensure performance, maintainability, and long-term scalability of the system.

Component	Technology	Notes
Frontend	JavaFX	Rich desktop UI framework for Java
Backend Logic	Java 17+	LTS release ensuring long-term support
Database	MySQL / MariaDB	Relational database for structured data storage
Reporting	JasperReports	Enterprise-grade report generation and PDF output
Build Tool	Maven	Dependency management and build automation
PDF Generation	JasperReports	Integrated PDF export from report engine
Authentication	Database-based	User credentials stored with hashed passwords
Backup	MySQL Dump + Java	Automated and manual database backup mechanism

4. Functional Requirements

4.1 Authentication & Authorization

Features

- User login and logout
- Change password (self-service)
- Reset password (Admin privilege)
- Role-based access control (RBAC)
- Permission-based module access

Roles & Access

Role	Access Level
Admin	Full system access including user management, cancellations, and restore
User	Limited access to authorized modules only

4.2 Region Management

Features

- Create region
- Edit region
- View all regions
- Activate / Deactivate regions

Region Fields

Field	Description
Region Code	Unique alphanumeric identifier for the region
Region Name	Full descriptive name of the region
Status	Active or Inactive

4.3 Church Management

Features

- Create church
- Edit church
- View all churches
- Activate / Deactivate churches

Church Fields

Field	Description
Church Code	Unique alphanumeric identifier for the church
Church Name	Full name of the church
Region	Linked parent region
Status	Active or Inactive

4.4 Receipt Management

> *Receipts are the core transactional records of the system. Each receipt represents a weekly collection submission from a church.*

Features

- Create weekly submission receipt
- View receipt history
- Cancel receipt (Admin permission required)
- Create corrected replacement receipt
- View submission status

Receipt Number Format

```
REC + YY + Running Number (resets annually)
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Example	Explanation
REC26000001	2026 — First receipt
REC26000002	2026 — Second receipt
REC27000001	2027 — Running number resets to 1

Business Rules

- No receipt reprints allowed
- One active receipt per church per week
- Week is defined as Monday to Sunday
- Collections submitted every Monday for the previous week
- Receipt may contain multiple collection items
- Duplicate collection types are not allowed within the same receipt

Collection Types

Type	Description
Offertory	Regular weekly offerings from church members
Tithes	Ten percent giving from members

Type	Description
Other Donations	Additional contributions and special offerings

Receipt Fields

Field	Description
Receipt No	Auto-generated using format REC+YY+RunNum
Church	Selected from active church list
Church Code	Auto-filled based on church selection
Region	Auto-filled based on church selection
Week Start Date	Monday of the submission week
Week End Date	Sunday of the submission week
Receipt DateTime	System-generated timestamp
Bearer Name	Manual text input
Submitted By Name	Manual text input
Issued By	Logged-in user (auto-filled)
Status	Active or Cancelled

4.5 Receipt Cancellation

> *Cancellation ensures data integrity — receipts cannot be edited; instead they are cancelled and replaced with a corrected record.*

Rules

- Receipts cannot be directly edited after issuance
- Admin permission is required to cancel a receipt
- Cancellation reason is mandatory
- A new corrected receipt must be created after cancellation
- New receipt is linked to the cancelled receipt for traceability

4.6 Reports

Weekly Reports

- Church-wise weekly summary
- Region-wise weekly summary
- Submission status report

Annual Reports

- Annual collection report

Progress Reports

- Church progress report

Additional Reports

- Missing submission report
- Cancelled receipt report
- Collection type summary
- User activity report
- Daily receipt report

4.7 Activity Logging

All major system activities must be logged to support security auditing and compliance.

Logged Activities

- Login and logout events
- Password changes
- Receipt creation and cancellation
- User account creation and updates
- Backup and restore operations

Log Record Fields

Field	Description
User	Username performing the action
Action	Type of activity performed
Module	System module where action occurred
Record ID	Reference to the affected record
Old Value	Data before the change
New Value	Data after the change
Machine Name	Name of the device used
Timestamp	Exact date and time of the action

4.8 Backup & Restore

Mode	Description
Manual Backup	User selects target folder; backup generated on demand
Automatic Backup	Scheduled backups with configurable time, folder, and retention policy
Restore	Admin-only; requires confirmation; auto-backup created before any restore

4.9 Submission Status Tracking

The system must provide real-time visibility into the weekly submission status:

- Churches that have submitted
- Churches with pending submissions
- Cancelled submissions
- Overall weekly submission completion percentage

5. Non-Functional Requirements

Quality Attribute	Requirements
Performance	Receipt generation must complete under 2 seconds. Reports generated with minimal latency. Support at least 10 concurrent LAN users in future deployment.
Security	Password hashing for all stored credentials. Role-based permissions enforced at service layer. Activity audit logs for all sensitive operations. Input validation and SQL injection prevention throughout. Transaction management with rollback support.
Reliability	Automatic scheduled backups to prevent data loss. Restore mechanism with pre-restore safety backup. Transaction rollback on operation failure.
Maintainability	Strict layered architecture with clear separation of concerns. UI logic separated from business and database logic. Modular code structure for ease of change.
Usability	Simple and intuitive user interface. User-friendly navigation with minimal training required.

6. System Architecture

6.1 Layered Architecture

The system follows a strict four-layer architecture:

JavaFX UI Layer	Presentation layer — forms, views, navigation
Service Layer	Business logic, validation, transaction management
Repository / DAO Layer	Data access abstraction, CRUD operations
MySQL / MariaDB	Persistent relational data storage

6.2 Internal Module Structure

Module	Responsibility
config	Application configuration and properties
controller	JavaFX controller classes for UI interaction
dto	Data Transfer Objects for inter-layer communication
entity	JPA/database entity classes
repository	DAO / Repository classes for data access
service	Business logic and service classes
security	Authentication, authorization, session management
reports	JasperReports definitions and report utilities
util	Common utility and helper classes
validation	Input validation and business rule enforcement

7. Database Requirements

7.1 Main Tables

Table	Purpose
regions	Stores region records — code, name, status
churches	Stores church records linked to regions
users	User accounts with hashed passwords
roles	System roles (Admin, User, etc.)
permissions	Granular permission definitions
role_permissions	Many-to-many mapping of roles to permissions
receipts	Weekly collection receipt headers
receipt_items	Line items per receipt (collection type + amount)
receipt_cancellations	Cancellation records linked to original receipts
activity_logs	Full audit log of system activities
backup_logs	Records of backup and restore operations

7.2 Key Constraints

- One active receipt per church per week (unique constraint)
- Unique receipt numbers within each year
- No duplicate collection types within the same receipt
- Foreign key relationships enforced between receipts, churches, and regions
- Cancellations linked to original receipt via foreign key

8. Future Enhancements

> *The system architecture has been designed to accommodate the following planned enhancements with minimal refactoring.*

- Web application front-end (browser-based access)
- Mobile application (iOS and Android)
- Cloud deployment for remote access
- Multi-branch remote access over the internet
- REST API integrations with third-party systems
- Real-time dashboard analytics
- Advanced reporting with export to Excel / CSV

9. Deployment Requirements

Phase	OS Requirement	Configuration
Initial Deployment	Windows 10 or above	Single machine, standalone
Future Deployment	Windows Server / Linux	Multiple JavaFX clients, centralized DB, LAN

10. Conclusion

The Church Collection Management System is designed as a scalable, secure, maintainable, and user-friendly desktop application capable of handling weekly collection management operations for multiple churches and regions.

The system architecture — built on JavaFX and MySQL with a strict layered pattern — supports future expansion into a centralized multi-user environment while maintaining simplicity for initial standalone deployment.

By adhering to the functional and non-functional requirements outlined in this document, the development team will deliver a robust solution that meets the operational, security, and reporting needs of the organization.

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